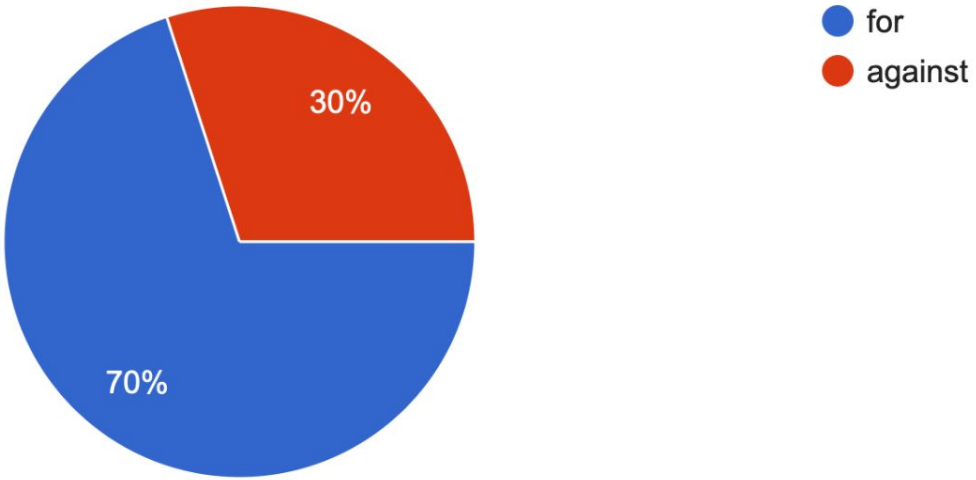
THE MONITOR DRONE

By the nutty ninjas

Opinion

Are you for the monitor drone

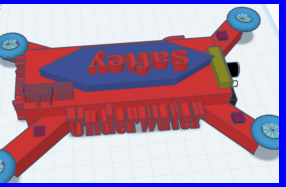
10 responses



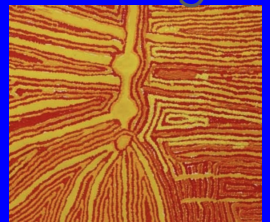
The Project:

The monitor drone is an innovative and sustainable drone.It blend seamlessly into the environment using eco friendly material like recycled oak wood ,and bamboo.It is also adorned with artwork from the the local artists of burnong descent appealing to a wider community,and to raise awareness about larger ocean conversation.The drone is built to be durable and weather resistant,and affordable for the oceans and all people.After surveying beachgoers 70% were in favour of this drone.The drone also has in built cameras.The drone is designed to monitor the seas for plastics and report it to its owners to better survey and safeguard the seas from all types of dangerous plastics.

Drone model



Burnong art



Solar panel

The solar panels attached to the drone will power the drone by charging on the sunlight in the sky.The satellite will be made of bamboo,recyclable plastic,circuits.When there is no sunlight the drone will be powered by reserve engines stored underneath.The solar panel on the drone being a solar panel has a strong glass beginning that it can survive contact with sea water.

Solar panels on houses used to provide an energy

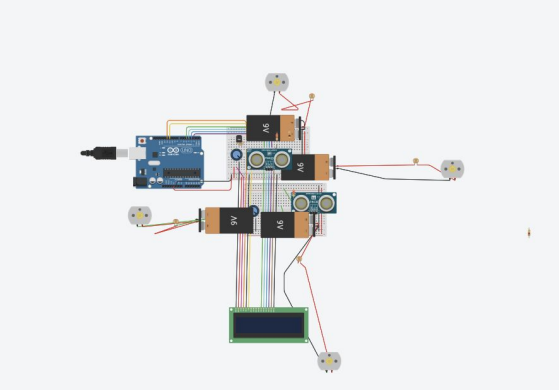
The small purple squares represent the solar panels

Circuits

The circuits attached to the drone will help the drone function smoothly doing its operations.The circuits will be shipped from the people's republics of china ensuring its stable costing.

example of circuits

Model of what circuits would look like in a real first real life model the V9 represents the reserve engines.The model excludes the camera and satellites

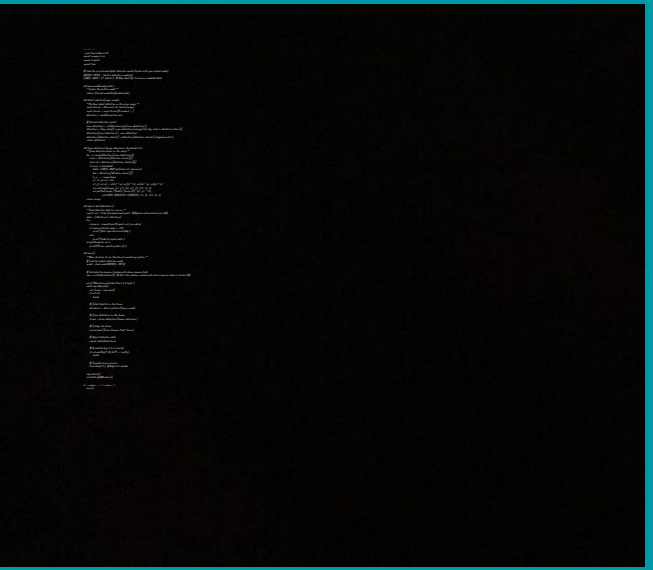


example of circuits

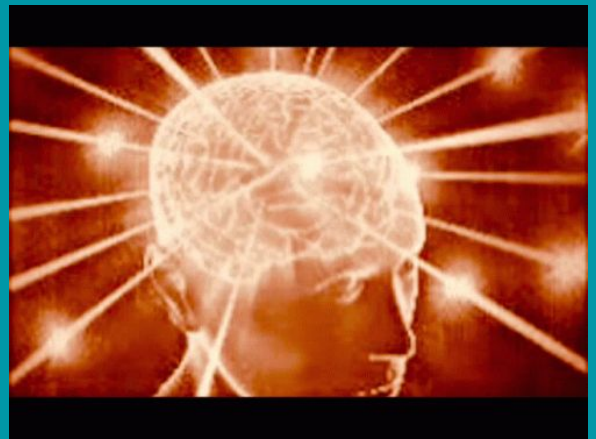
Coding

There is a coding element also present in the innovation project.The coding will be connected to the robot by the circuits.This is a example on what the coding would look like.The coding aspect with the help of artificial intelligence will cost nothing.The coding can also be shaped with a.i to withstand the oceans and keep up with the demand of the great numbers of plastics in the oceans.We test our python coding with computer programs to make sure the code can be uploaded to the drones.

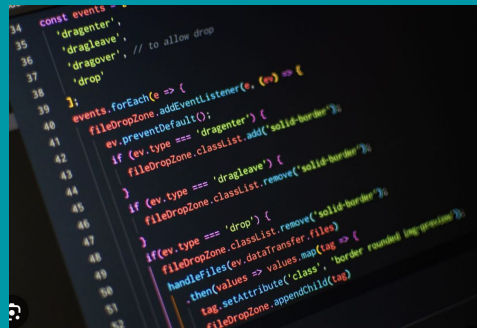
python coding



gif of a person being smart



example of code



Reserve engines

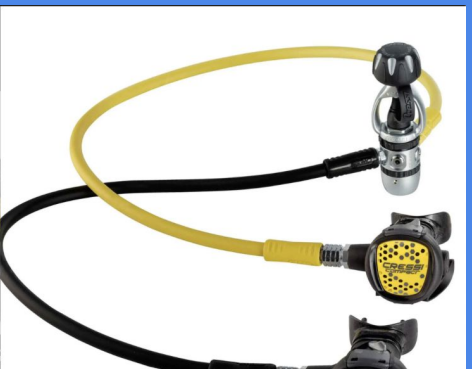
The drone will rely on rotary combustion engines as a backup engine if the drone satellites can't directly access sunlight .A rotary combustion engine is a type of complex internal engine that uses motors to move the drone.Based on the measurements of the drone the measurements of the rotary combustion engines would be quiet small it weighs less than 1kg based on company of manufacturing input.It has a power of around 1 to 10 HP.Serving as a vital backup in a marine monetisation environment.

Rotary combustion engine

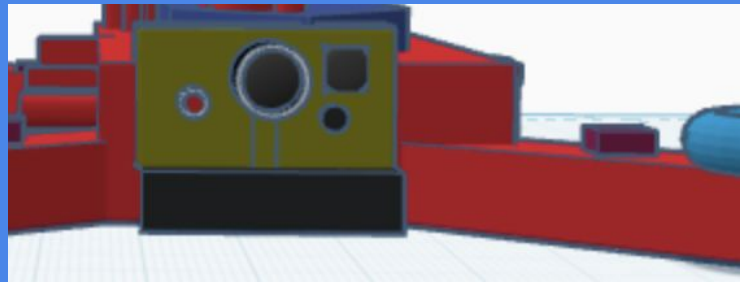
Built in Camera

The built in camera is a camera that is built in from the drone.Using the rotary combustion engines as a basis we power the camera using a rotary combustion regulator.The built in camera is inspired from the cameras in the early 1990's because of the decades impact of human technology and geopolitics.

Rotary combustion regulator



Camera attached to drone



Intergration process

The goal of the monitor drone is to not dominate the marine monetisation but to serve as to a premise to better international models. The monitor drone would be updated roughly 2 weeks after release and the design blueprints without copyright(that do not exist because this is just an innovation project) would go out to the general public to create many different and unique models to this monitor drone and to encourage international competition to make the best versions for low prices.The drones would be manufactured in shanghai to ensure feasibility.

blueprint design example

Proof of concept

The concept of the monitor drone is to detect plastics in the oceans for the cheapest price of just $5 USD to stop marine,and human life loss due to excessive amounts of plastics in the ocean.The data recorded from the drone from the cameras would be sent to detection logs of the owner of the drone.The drone would be pre made and set up in the P.R.C for immediate shipping without need of assembly.

Blueprint design example

Measurements & costs

Approximate measurements

Drone

height(30mm)

width(115mm)

Length(170mm)

Rotary Combustion Engine

height(30mm)

width(115mm)

Length(170mm)

Rotary Combustion regulator

Height(20mm)

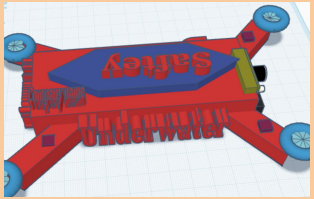
Width(30mm)

Length(50mm)

weight(30 grams)

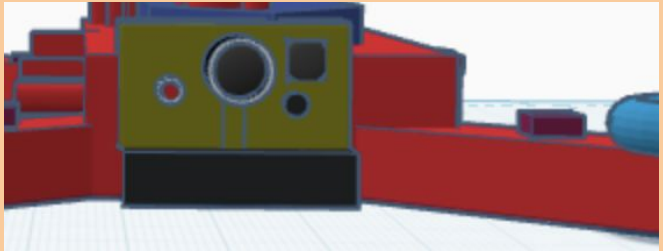
camera

drone model



Plz note :finish of camera measurements when 3d printed

Rotary Combustion Engines



Camera and Drone Model

Outreach

As a part of our project we conducted a sophisticated survey of beachgoers to understand their attitudes towards innovative drones affecting the waters.We found that 70% of people would approve with this drone product.This feedback confirms that our design resonates with the public and their needs of a practical and effective drone monitoring the seas so that the general public will not be affected by plastic.

Picture of beach

The problem we are solving

International pollution spread by the sea is a problem and a threat to humanity.Our monitor drone will fix those problems and fix waste disposals by surveying the plastics ensuring proper authorities and beachgoers dispose of plastics ending the international crisis.The goal of the monitor drone is to not dominate the marine monetisation but to serve as to a premise to better international models.

Picture of polluted becah

Locations

The manufacturing and design scale is of international level feasibility is of the cheap costs and profitability of multiple company and civilians to earn money

Bamboo-From People's Republic Of China

Circuits-From People's Republic Of China

Oak-North east of victoria located in the Commonwealth of Australian government

Burnong-Victoria located in the north east of the Commonwealth of Australian government

P.R.C

Victoria

Compass

sources

References:

SkyX. (2023). *Drone Technology for Protecting Marine Life and Ecosystems*. Retrieved from https://skyx.com/blog/when-sky-meets-sea-drone-technology-for-protecting-marine-life-and-ecosystems/

https://pangeaseed.org/project/sea-walls-artists-for-oceans/

Washed Ashore. (2023). *Art to Save the Sea*. Retrieved from https://www.washedashore.org/

NOAA. (2023). *These 5 Technologies Are Helping Save Our Ocean*. Retrieved from https://www.noaa.gov/stories/these-5-technologies-are-helping-save-our-ocean

Living Oceans Foundation. (2023). *Inspiring Ocean Conservation Through Art*. Retrieved from https://www.livingoceansfoundation.org/publication/inspiring-ocean-conservation-through-art

https://www.nature.com/articles/s41598-021-83477-6Pucino, N., Kennedy, D. M., Carvalho, R. C., Allan, B., & Ierodiaconou, D. (2021). Citizen science for monitoring seasonal-scale beach erosion and behaviour with aerial drones. *Scientific reports*, *11*(1), 3935.

https://www.unmannedsystemstechnology.com/expo/military-drone-engines/#:~:text=Military%20drone%20engines%20may%20utilize,driven%20and%20rotary%20Wankel%20engines

https://docs.google.com/forms/d/e/1FAIpQLSfWi8iHVMDWD7mtNcXXon_EXY-VoHsUasbcGY9Eu_mMVyQg/viewform?pli=1&pli=1 survey

Inuja jayasinghe.(2024).Funky bruticus.Retrieved from safari

https://www.tinkercad.com/things/k9L41R/2o2k-funky-bruticus/edit?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard

https://www.britannica.com/place/China

https://en.wikipedia.org/wiki/Geography_of_Victoria

https://www.amazon.com.au/Nautical-Vintage-Antique-Finish-Compass/dp/B00W5IYO9I

https://info.pennthaplar.com.au/welcome?SOF_SOURCE=GoogleAds&kw=solar%20panel&cpn=22046394922&utm_term=solar%20panel&utm_campaign=Tesla+Powerwall+%7C+Search&utm_source=adwords&utm_medium=ppc&hsa_acc=4870398906&hsa_cam=22046394922&hsa_grp=171888562266&hsa_ad=725950668873&hsa_src=g&hsa_tgt=kwd-116542991&hsa_kw=solar%20panel&hsa_mtr=e&hsa_net=adwords&hsa_ver=3&gad_source=1&gbraid=0AAAAACF9Yb7QvE6B9ZBkxzOJpxKLk7677Z&gclid=Cj0KCQjAutmTBhC5ARIsAFiwNHvXcYOYNv3IuwOdUZ_Hc7Bcr_fqO3uo1bTivvFxoNJxt_Ad9LCh-MqAs2IrEALw_wcB

https://www.enrgtech.co.uk/blog/how-does-a-dedicated-computer-circuit-work-for-home-offices/

https://www.mtek.co.uk/what-is-a-computer-circuit-board

https://www.blueprintcd.com.au/

Troubleshooting

1-The image was intended to be a walking drone but could not be found hence why the image was of an family with an drone.

2-It took a long time to get survey results because of focus on more complex issues

3-We had to copy and paste the drone 3d model running the risk of being copied but after applied feedback from the team head of the innovation project of “It is a low chance of being copied” we applied the 3d drone model

4-No issues or feedback of improvement

5-It was to complex to scale the whole drone circuitry so we had to scale out the satellite ,reverse engines,and rotary combustion engines

6-It took a long time to get appropriate coding format but from the head of innovation project feedback we managed to get the correct format of python coding.

7-No issues or feedback of improvement

8-No issues or feedback of improvement

9-No issues or feedback of improvement

10-No issues or feedback of improvement

11-No issues or feedback of improvement

12-No issues or feedback of improvement

13-No issues or feedback of improvement

14-It was very complicated to get accurate photos on locations apart from wikipedia and britannica

15-No issues or feedback of improvement

Dates Made

The slides were made in 1st of April to 2nd of April